Information retrieval: precision and recall

Information retrieval is the task of finding material in an unstructured store that matches an information need.

Precision measures the fraction of retrieved documents that are relevant. Recall measures the fraction of relevant documents that were retrieved. The two pull in opposite directions: returning more documents raises recall and lowers precision.

A ranked retrieval system returns an ordered list, so a cutoff at rank k gives a precision-at-k figure. The break-even point is where precision equals recall.

Relevance scoring

TF-IDF weights a term by how often it appears in a document and how rare it is across the collection. A term appearing in every document carries little discriminative weight.

BM25 is a probabilistic ranking function that extends TF-IDF with term-frequency saturation and document-length normalisation. Saturation means that the tenth occurrence of a word adds much less than the second.

The k1 and b parameters control saturation and length normalisation. With b set to 0, document length is ignored entirely.

Evaluation

Evaluation uses a test collection of documents, queries and a judgement of which documents are relevant.

Mean average precision averages precision at each relevant retrieved position over all queries. Normalised discounted cumulative gain accounts for graded rather than binary relevance.

An inverted index maps each term to the postings list of documents that contain it, which is what makes ranked retrieval fast enough to serve interactively.